V4B5 - Real and Harmonic Analysis

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1 L^p spaces

1.1 Basic Measure Theory

Notation 1.1. $\mathbb{T} = \{z \in \mathbb{C} \mid z = 1\}$.

Notation 1.2. *For $n \in \mathbb{Z}$, $e_n : \mathbb{T} \to \mathbb{C}$ is*

$$e_n(x) = \exp(2\pi i n x).$$

Definition 1.1. *$L^2(\mathbb{T})$ is a Hilbert space with an inner product such that for $f, g \in L^2(\mathbb{T})$,*

$$\langle f, g \rangle = \int_{\mathbb{T}} f(x)\overline{g(x)}dx.$$

Definition 1.2. *For $f \in L^2(\mathbb{T})$, we define a map $\hat{f} : \mathbb{Z} \to \mathbb{C}$ such that*

$$\hat{f}(n) := \langle f, e_n \rangle.$$

Definition 1.3. *$\mathcal{M}(\mathbb{T})$ denote the set of all finite complex Borel measures on $\mathbb{T}$. The 1-norm is defined by*

$$\|\mu\|_{\mathcal{M}(\mathbb{T})} = \sup_{\{I_n\} \in \mathcal{I}} \bigcup_n |\mu(I_n)|,$$

where $\mathcal{I}$ is the collection of sets of subsets in $\mathbb{T}$ pairwise disjoint.

Proposition 1.1. *We have a map*

$$L^p(\mathbb{T}) \ni f \mapsto f(x)dx \in \mathcal{M}(\mathbb{T}),$$

which is injective.

Notation 1.3. *For $\mu \in \mathcal{M}(\mathbb{T})$, we set,*

$$\hat{\mu}(n) := \langle \mu, e_n \rangle = \int_0^1 \exp(-2nx\pi i)d\mu(x).$$

Proposition 1.2. *The collection $\{e_k\}_{k\in\mathbb{Z}}$ is an orthonormal set in $L^2(\mathbb{T})$ with Lebesgue measure.*

Proof.

$$\begin{aligned}\langle e_j, e_k \rangle &= \int_0^1 \exp(2\pi i(j-k)x)dx, \\ &= \begin{cases} 1, & j = k, \\ \frac{\exp(2\pi i(j-k)x)}{2\pi(j-k)}\Big|_0^1 = 0, & j \neq k. \end{cases}\end{aligned}$$

□

Definition 1.4. *A trigonometric polynomial is a finite linear combination of $\{e_k\}_{k\in\mathbb{Z}}$.*

Corollary 1.1. *Suppose f is a trigonometric polynomial then we have,*

$$f(x) = \sum_{k\in\mathbb{Z}} \hat{f}(x)e_k(x).$$

which is pointwise on $\mathbb{T}$.

Proof. Follows the orthonormality from Proposition 1.2 □

Definition 1.5. *Let $f \in L^1(\mathbb{T})$ and $N \in \mathbb{N}$, we define,*

$$S_N f(x) := \sum_{\substack{k\in\mathbb{Z}\\ |k|\leq N}} \hat{f}(k)e_k(x).$$

Definition 1.6. *For $N \in \mathbb{N}$, the Dirichlet kernel $D_N : [0,1] \to \mathbb{C}$ is*

$$D_N(x) = \sum_{\substack{k\in\mathbb{Z}\\ |k|\leq N}} e_k(x).$$

Definition 1.7. *For $f, g \in L^1(\mathbb{T})$, the convolution $f * g$ is*

$$f * g(x) = \int_0^1 f(x-y)g(y)dy.$$

Proposition 1.3. *For $f \in L^1(\mathbb{T})$,*

$$S_N f = D_N * f.$$

Proof.

$$\begin{aligned} S_N f(x) &= \sum_{|k|\leq N} \hat{f}(k)e_k(x), \\ &= \sum_{|k|\leq N} \int_0^1 f(y)e_k(-y)dy \cdot e_k(x), \\ &= \int_0^1 f(y) \sum_{|k|\leq N} e_k(x-y)dy, \\ &= \int_0^1 f(y)D_N(x-y). \end{aligned}$$

□

Notation 1.4. *$a \lesssim b$ denotes that there is a constant $C > 0$ such that $a \leq Cb$.*

Proposition 1.4.

$$|D_N(x)| \lesssim \min\left\{N, \frac{1}{x}\right\}.$$

Proof. Observe that

$$D_N = e_{-N}\frac{1-e_{2N+1}}{1-e_1} = \frac{e_{N+\frac{1}{2}} - e_{-N-\frac{1}{2}}}{e_{\frac{1}{2}} - e_{-\frac{1}{2}}} = \frac{\sin((2N+1)\pi x)}{\sin(\pi x)}.$$

Since the numerator is bounded by 1 and $\sin(\pi x)$ is bounded by some constant multiple of $\frac{1}{|x|}$ around the origin where $\mathbb{T} \cong \left[-\frac{1}{2}, \frac{1}{2}\right]$. Also observe that

$$|D_N| \leq \sum_{|k|\leq N} |e_k| = N + 1.$$

□

Proposition 1.5.

$$(f * g)(x - t) = (f * g(\cdot - t))(x) = (f(\cdot - x)*)(x).$$

In other words, the convolution commutes with the translation.

Proof.

$$\begin{aligned}(f * g)(x - t) &= \int_0^1 f(x - y - t)g(y)dy, \\ &= \int_0^1 f(x - y)g(y - t)dy, \\ &= (f * g(\cdot - t))(x).\end{aligned}$$

□

Proposition 1.6 (Fubini)**.** *Suppose $F \in L^1(X \times Y)$, then*

$$\int_Y \int_X F(x, y)dxdy = \int_X \int_Y F(x, y)dydx.$$

Proposition 1.7. *Let $f, g \in L^1(\mathbb{T})$, then $f * g \in L^1(\mathbb{T})$.*

$$\begin{aligned}\int_{\mathbb{T}} |f * g(x)|dx &\leq \int_{\mathbb{T}} \int_{\mathbb{T}} |f(x - y)g(y)|dydx, \\ &\overset{\text{Fubini}}{=} \int_{\mathbb{T}} |g(y)| \int_{\mathbb{T}} |f(x - y)|dxdy, \\ &= \|f\|_{L^1} \|g\|_{L^1}.\end{aligned}$$

Theorem 1.1 (Minkowski's Inequality)**.** *Let X be a measure space and $1 \leq p \leq \infty$. Then for any $f, g \in L^p(X)$, we have*

$$\|f + g\|_{L^p(X)} \leq \|f\|_{L^p(X)} + \|g\|_{L^p(X)}.$$

In particular $L^p(X)$ is closed under addition.

Theorem 1.2 (Hölder's Inequality)**.** *Let X be a measure space and $1 \leq p, q \leq \infty$ be such that*

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then for any $f \in L^p(X), g \in L^q(X)$, we have

$$\|fg\|_{L^1(X)} \leq \|f\|_{L^p(X)} \|g\|_{L^q(X)}.$$

Theorem 1.3 (Riesz-Thorin Interpolation)**.** *Let X, Y be measure spaces and T be a linear map between measurable functionals on X to measurable functionals on Y. Suppose for some $p_1, q_1, p_2, q_2 \in [1, \infty]$, we have,*

$$T(L^{p_1}(X)) \subseteq L^{q_1}(Y), \quad T(L^{p_2}(X)) \subseteq L^{q_2}(Y).$$

Then for any $\theta \in [0, 1]$, set (p, q) to be

$$\frac{1}{p} = \frac{\theta}{p_1} + \frac{1 - \theta}{p_2}, \quad \frac{1}{q} = \frac{\theta}{q_1} + \frac{1 - \theta}{q_2},$$

then

$$T(L^p(X)) \subseteq L^q(Y).$$

In particular,

$$\|T\|_{L^p(X) \to L^q(Y)} \leq \|T\|^{\theta}_{L^{p_1}(X) \to L^{q_1}(Y)} \|T\|^{1-\theta}_{L^{p_2}(X) \to L^{q_2}(Y)}.$$

Definition 1.8. *Let $f : X \to \mathbb{R}$ be a functional on a measurable space.*

$$\text{esssup}_{x\in X}\, f(x) := \{M \in \mathbb{R} \mid (x) \leq M \text{ almost everywhere.}\}.$$

Theorem 1.4 (Young's Inequality)**.** *Let $f \in L^p(\mathbb{T}), g \in L^q(\mathbb{T})$. Then for*

$$1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q},$$

we have,

$$\|f * g\|_{L^r(\mathbb{T})} \leq \|f\|_{L^p(\mathbb{T})}\|g\|_{L^q(\mathbb{T})}.$$

Proof. Fix p and we vary q.

Case 1 : $q = 1$, therefore $r = p$.

$$\begin{aligned}
\|f * g\|_{L^p(\mathbb{T})} &= \left\| \int_{\mathbb{T}} f(x-y)g(y)dy \right\|_{L^p(\mathbb{T})}, \\
&\overset{\text{Minkowski}}{\leq} \int_{\mathbb{T}} \|f(x-y)g(y)\|_{L^p(\mathbb{T})} dy, \\
&= \|f(x-y)\|_{L^p(\mathbb{T})} \int_{\mathbb{T}} |g(y)|dy, \\
&= \|f\|_{L^p(\mathbb{T})}\|g\|_{L^1(\mathbb{T})}.
\end{aligned}$$

Case 2 : $q = p'$ where $p' = \frac{p}{1-p}$ so that $r = \infty$. Therefore,

$$\begin{aligned}
\|f * g\|_{L^\infty(\mathbb{T})} &= \text{esssup}_{x\in\mathbb{T}} \left| \int_{\mathbb{T}} f(x-y)g(y)dy \right|, \\
&\overset{\text{Hölder}}{\leq} \|f\|_{L^p(\mathbb{T})}\|g\|_{L^q(\mathbb{T})}.
\end{aligned}$$

General Case, for fixed $f \in L^p(\mathbb{T})$, consider the linear map

$$L^q(\mathbb{T}) \ni g \overset{T}{\mapsto} f * g \in L^r(\mathbb{T}).$$

Notice that we have that T is bounded for $(r, q) = (p, 1), (\infty, p')$ cases in Theorem 1.3. Furthermore, for $\theta \in [0, 1]$, set

$$\begin{aligned}
&\frac{1}{q} = \theta + (1-\theta)\frac{(p-1)}{p} = 1 - (1-\theta)\frac{1}{p}, \\
&\frac{1}{r} = \theta\frac{1}{p}.
\end{aligned}$$

Then

$$\frac{1}{r} - \frac{1}{q} = 1 - \frac{1}{p}.$$

Therefore, the statement is shown. □

Definition 1.9. *Let $f \in C^0(\mathbb{T})$ and $\mu \in \mathcal{M}(\mathbb{T})$ then $f * \mu$ is defined as,*

$$f * \mu(x) = \int_{\mathbb{T}} (f(x-y))d\mu(y).$$

Corollary 1.2. *In Definition 1.9, $f * \mu$ is well-defined.*

Proof. We need to show that

$$\|f * \mu\|_{L^p(\mathbb{T})} \leq \|f\|_{L^p(\mathbb{T})}\|\mu\|_{\mathcal{M}(\mathbb{T})}.$$

This follows from $q = 1$ case in THeorem 1.4. □

Corollary 1.3. *Let $f \in L^p(\mathbb{T})$ and $g \in L^q(\mathbb{T})$ be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then $f * g$ is continuous.*

Proof. Since $\cdot * g$ is linear together with Proposition 1.5, we have,

$$(f * g)(x + h) - (f * g)(x) = ((f(\cdot + h) - f) * g)(x).$$

Therefore by Hölder,

$$|((f(\cdot + h) - f) * g)(x)| \leq \|f(\cdot + h) - f\|_{L^p(\mathbb{T})} \|g\|_{L^q(\mathbb{T})}.$$

As $\|f(\cdot + h) - f\|_{L^p(\mathbb{T})} \overset{h \to 0}{\to} 0$, we have shown the statement. □

Lemma 1.1. *For $f, g \in L^1(\mathbb{T})$, we have,*

$$\widehat{f * g}(n) = \hat{f}(n)\hat{g}(n).$$

Proof.

$$\begin{aligned}
\widehat{f * g}(n) &= \int_{\mathbb{T}} (f * g)(x) e_n(-x) dx, \\
&= \int_{\mathbb{T}} \int_{\mathbb{T}} f(x - y) g(y) dy e_n(-x) dx, \\
&= \int_{\mathbb{T}} \int_{\mathbb{T}} f(x - y) e_n(-(x - y)) g(y) e_n(-x) dy dx, \\
&\overset{\text{Fubini}}{=} \int_{\mathbb{T}} \int_{\mathbb{T}} f(x - y) e_n(-(x - y)) g(y) e_n(-x) dx dy, \\
&= \hat{f}(n)\hat{g}(n).
\end{aligned}$$

□

We now extend the application of Theorem 1.3 to non-linear case.

Definition 1.10. *Let X, Y be topological vectorspaces and $D \subseteq X$ be a dense subset. An operator $T : D \to Y$ is said to be sublinear if for any $x, y \in D, c \in \mathbb{R}$, we have,*

$$\begin{aligned}
|T(x + y)| &\leq |T(x)| + |T(y)|, \\
|T(cx)| &= |c||T(x)|.
\end{aligned}$$

Definition 1.11. *For $p \in [1, \infty]$ and a measure space (X, μ), a functional $f : X \to \mathbb{R}$ its weak L^p-norm is*

$$\|f\|_{L^{p,\infty}(X,\mu)} = \sup_{\lambda \in (0,\infty)} \lambda \mu(\{x \in X \mid f(x) > \lambda\})^{\frac{1}{p}}.$$

The weak L^p-space on (X, μ) is a set of all functions $f : X \to \mathbb{R}$ whose weak L^p-norm is finite.

Remark 1.1. $L^{\infty,\infty}(X, \mu) = L^\infty(X, \mu)$

Proposition 1.8. *For any $p \in [1, \infty]$, we have*

$$L^p(X) \subseteq L^{p,\infty}(X).$$

Proof.

$$\begin{aligned}
\mu(\{x \in X \mid f(x) > \lambda\}) &\leq \int_{\{x \in X \mid f(x) > \lambda\}} \frac{|f(x)|^p}{\lambda} d\mu(x), \\
&\leq \frac{\|f\|^p_{L^p(X)}}{\lambda^p}.
\end{aligned}$$

□

Notation 1.5. *For measurable spaces $X, \tilde{X}$, let T be a sublinear operator sending functionals on X to functionals on $\tilde{X}$. We denote,*

$$T : L^p(X) \to L^{q,\infty}(\tilde{X})$$

if there is $c > 0$ such that for the dense subset $\mathcal{D}$ of $L^p(X)$ which T is defined on

$$\forall f \in \mathcal{D}, \|Tf\|_{L^{q,\infty}(\tilde{X})} \leq c\|f\|_{L^p(X)}.$$

Theorem 1.5 (Marcinkiewicz's Interpolation)**.** *For measurable spaces $X, \tilde{X}$, let T be a sublinear operator sending functionals on X to functionals on $\tilde{X}$ and $p_1, p_2, q_1, q_2 \in [1, \infty)$ such that $q_1 \neq q_2$ and*

$$\frac{1}{p_1} \geq \frac{1}{q_1}, \frac{1}{p_2} \geq \frac{1}{q_2}.$$

Suppose we have

$$T : L^{p_1}(X) \to L^{q_1,\infty}(\tilde{X}), T : L^{p_2}(X) \to L^{q_2,\infty}(\tilde{X}),$$

then for any $\theta \in (0, 1)$ and a pair (p, q) such that

$$\left(\frac{1}{p}, \frac{1}{q}\right) = \theta\left(\frac{1}{p_1}, \frac{1}{q_1}\right) + (1-\theta)\left(\frac{1}{p_2}, \frac{1}{q_2}\right),$$

we have

$$T : L^p(X) \to L^q(\tilde{X}).$$

Remark 1.2. *In applications, we will encounter operators such that $T : L^p(X) \to L^p(\tilde{X})$ for $p \in (1, \infty)$ but fails at $p = 1$. The above theorem is often applied after showing $T : L^1(X) \to L^{1,\infty}(\tilde{X})$ and $T : L^\infty(X) \to L^\infty(\tilde{X})$.*

1.2 Fourier Transform

Lemma 1.2. *For a differentiable function $f : \mathbb{T} \to \mathbb{R}$, we have,*

$$\hat{f}'(n) = 2\pi i n \hat{f}(n).$$

Proof.

$$\int_{\mathbb{T}} f'(x)e_n(-x)dx = f(x)e_n(-x)|_0^1 + 2\pi i n \int_{\mathbb{T}} f(x)e_n(-x)dx.$$

□

Definition 1.12. *Let $f : \mathbb{Z} \to \mathbb{R}$ be a function then the discrete derivative of f is*

$$\Delta f(n) = f(n) - f(n-1).$$

Lemma 1.3. *Let $z = e^{2\pi i x}$ then,*

$$\widehat{(1-z)f}(n) = \Delta\hat{f}(n).$$

Lemma 1.4.

$$f \in \mathcal{C}^k(\mathbb{T}) \Rightarrow \exists C > 0 \text{ such that } \forall n \in \mathbb{Z} |\hat{f}(n)| \leq \frac{C}{n^k}.$$

Proof. For $k = 1$,

$$\begin{aligned}\hat{f}(n) &= \int_0^1 f(x)e_n(-x)dx, \\ &= f(x)\frac{e_n(-x)}{-2\pi i n}|_0^1 + \frac{1}{2\pi i n}\int_0^1 f'(x)e_n(-x)dx.\end{aligned}$$

Since f' is continuous thus bounded on $\mathbb{T}$. Therefore, C can be taken as $\int_0^1 |f(x)|dx$.

By induction on k, the statement follows. □

Definition 1.13. *For $k \in \mathbb{N}$, $l^k(\mathbb{Z})$ space is the set of sequences $(x_n)_{n\in\mathbb{Z}}$ of complex numbers indexed by $\mathbb{Z}$ such that*

$$\sum_{n\in\mathbb{Z}} |x_n|^k < \infty.$$

Lemma 1.5. *If $f : \mathbb{T} \to \mathbb{C}$ is bounded by,*

$$|f(x)| \leq \frac{C}{1 - e_1(x)},$$

then $\Delta\hat{f} \in l^1(\mathbb{Z})$.

Putting these altogether, we obtain the following theorem.

Theorem 1.6. *We have the following duality*

Physical side	$\overset{\widehat{(\cdot)}}{\longleftrightarrow}$	*Frequency side*
Convolution	$\longleftrightarrow$	*Multiplication*
Differentiation	$\longleftrightarrow$	*Multiplication by polynomials*
Multiplication by trigonometric polynomials	$\longleftrightarrow$	*Discrete differentiation*
Regularity	$\longleftrightarrow$	*Decay*
Decay	$\longleftrightarrow$	*Regularity*

Remark 1.3.

$$\widehat{S_N f} = \hat{f} \cdot \chi_{[-N,N]}.$$

Definition 1.14. *Let $f \in L^1(\mathbb{T})$. We define the Cesaro mean of f to be*

$$\sigma_N f(x) := \frac{1}{N} \sum_{k=1}^{N-1} S_k f(x).$$

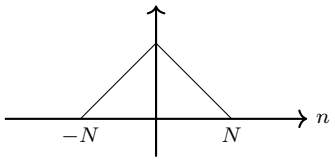
Observe that

$$\begin{aligned}\sigma_N f(x) &= \frac{1}{N} \sum_{k=0}^{N-1} (f * D_k)(x), \\ &= f * \left(\frac{1}{N} \sum_{k=0}^{N-1} D_k \right)(x).\end{aligned}$$

Definition 1.15. *Motivated by the above observation, we define the Fejer kernel to be,*

$$K_N := \frac{1}{N} \sum_{k=0}^{N-1} D_k.$$

Remark 1.4. *$\widehat{K_N}$ is graphed as*



Lemma 1.6. *For $N \in \mathbb{N}$,*

$$|\sin(Nx)| \leq N|\sin(x)|.$$

Proof. Induction on N using the addition theorem for trigonometric functions. □

Lemma 1.7. *There is $C > 0$ such that for any $x \in \mathbb{T}$,*

$$0 \leq K_N(x) \leq \frac{C}{N} \min\left\{N^2, \frac{1}{x^2}\right\}.$$

Proof. Let $z = e_1(x)$. From the proof of Proposition 1.4, we have

$$\begin{aligned} K_N(x)(z^{\frac{1}{2}} - z^{-\frac{1}{2}}) &= \frac{1}{N}\left(\sum_{k=0}^{N-1} z^{k+\frac{1}{2}} - z^{-k-\frac{1}{2}}\right), \\ &= \frac{1}{N}\left(\frac{z^N - 1}{z^{\frac{1}{2}} - z^{-\frac{1}{2}}} + \frac{1 - z^{-N}}{z^{\frac{1}{2}} - z^{-\frac{1}{2}}}\right), \\ &= \frac{1}{N}\frac{(z^{\frac{1}{N}} - z^{-\frac{1}{N}})^2}{z^{\frac{1}{2}} - z^{-\frac{1}{2}}}, \\ &= \frac{1}{N}\frac{\sin^2(N\pi x)}{\sin^2(\pi x)} \geq 0. \end{aligned}$$

The upperbound follows from Taylor expansion and Lemma 1.6. □

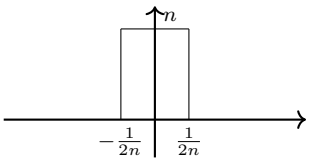
Definition 1.16. *A family of bounded complex-valued functions $\{\Phi_n\}_{n\in\mathbb{N}}$ on $\mathbb{T}$ is called an approximate identity if*

i. $\forall n \in \mathbb{N}, \int_{\mathbb{T}} \Phi_n(x)dx = 1.$

ii. $\sup_{n\in\mathbb{N}} \int_{\mathbb{T}} |\Phi_n(x)|dx < \infty.$

iii. $\forall \delta > 0, \lim_{n\to\infty} \int_{|x|>\delta} |\Phi_N(x)|dx = 0.$

Example 1.1. *Set $\Phi_n(x)$ to be*



$(\Phi_n)_{n\in\mathbb{N}}$ is called the box kernels. This is an approximate identity.

Theorem 1.7. *Fejér kernel is an approximate identity.*

Proof. Clearly $\int_{\mathbb{T}} K_n(x) = 1$ since for each D_k, the constant 1 term survives. Since K_n is everywhere non-negative, the second condition is shown immediately.

Lastly for $\delta > 0$, let $N > \frac{1}{\delta}$ then,

$$\begin{aligned} \int_{|y|>\delta} |K_N(y)|dy &\overset{\text{Lemma 1.7}}{\leq} 2C\int_{\delta}^{\frac{1}{2}} \frac{1}{Ny^2}dy, \\ &\leq \frac{1}{N}(\delta^{-1} - 2) \overset{N\to\infty}{\to} 0. \end{aligned}$$

□

Definition 1.17. *Let $(\mu_n)_{n\in\mathbb{N}}$ be a sequence in $\mathcal{M}(\mathbb{T})$. We say μ_n converges to μ in weak-$\star$-sense if for any $f \in C^0(\mathbb{T})$,*

$$\int_{\mathbb{T}} f(x)d\mu_n(x) \overset{n\to\infty}{\to} \int_{\mathbb{T}} f(x)d\mu(x).$$

Proposition 1.9. *For any approximate identity $\{\Phi_n\}_{n\in\mathbb{N}}$, the following statements hold.*

1. *If $f \in C^0(\mathbb{T})$, then $\|\Phi_n * f - f\|_{L^\infty(\mathbb{T})} \to 0$ as $n \to \infty$.*
2. *For $p \in [1,\infty)$ and $f \in L^p(\mathbb{T})$, we have $\|\Phi_n * f - f\|_{L^p(\mathbb{T})} \to 0$ as $n \to \infty$.*
3. *If $\mu \in \mathcal{M}(\mathbb{T})$, then $|mu * \Phi_n \to \mu$ in weak-$\star$-sense.*

Proof. □